

Algebra First-Year Exam, May 2018, Part 2

Answer four problems. Write your solutions clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

1. Let D be a positive integer such that $-D \equiv 1 \pmod{4}$. Define

$$\omega = \frac{1 + \sqrt{-D}}{2}, \quad R = \{a + b\omega : a, b \in \mathbb{Z}\}.$$

For $\alpha \in R$ define $N(\alpha) = \alpha\bar{\alpha}$, where $\bar{\alpha}$ is the complex conjugate of α .

- (a) Prove that R is a subring of the field $\mathbb{C}$ of complex numbers.
 - (b) Prove that for all $\alpha \in R$ we have $N(\alpha) \in \mathbb{Z}_{\geq 0}$.
 - (c) Prove that for all $\alpha, \beta \in R$ we have $N(\alpha\beta) = N(\alpha)N(\beta)$.
 - (d) Let $\alpha \in R$. Prove that α is a unit if and only if $N(\alpha) = 1$.
2. This problem has two parts.
- (a) Let F be a field and let G be a finite subgroup of the unit group $F^\times$. Prove that G is cyclic.
 - (b) Give an example of a field F and a finitely generated subgroup G of $F^\times$ which is not cyclic.
3. Let R be a commutative ring with $1 \neq 0$ and let M be a unital R -module. Say M is irreducible if $M \neq \{0\}$ and the only submodules of M are $\{0\}$ and M . Prove that if M is an irreducible R -module then $M \cong R/I$ for some maximal ideal I in R .
4. Let V be a vector space over a field F and let S be a subset of V such that $\text{Span}(S) = V$. Use Zorn's lemma to prove that there is a subset $\mathcal{B}$ of S which is a basis for V .
5. Find a representative for each similarity class of 3×3 matrices A with entries in $\mathbb{F}_3 = \mathbb{Z}/3\mathbb{Z}$ such that $A^4 = A$. (Note that $X^4 - X$ factors into linear factors over $\mathbb{F}_3$.)
6. Let $\gamma \in \mathbb{C}$ be a root of $X^3 - 6X^2 + 2 \in \mathbb{Q}[X]$. Express $(1 + \gamma^2)(3 - 4\gamma^2)$ and $1/(1 + \gamma^2)$ in the form $a + b\gamma + c\gamma^2$, with $a, b, c \in \mathbb{Q}$.